

Key - for Problems not in Tutorials.

①

Question Bank.

② $R_{ab} = 142.32 \Omega$

④ $R_{ab} = 5.818 \Omega$

②b $V_1 = 30V$ $V_2 = 60V$ $V_3 = 60V$

③ $I_0 = I_3 = 1.7778A$ $V_{ab} = 53.33V$

④ $I_{\text{ammeter}} = 99.99 \mu A$

⑤ $R_{\text{max}} = 1k\Omega$ $3mW$.

⑤
$$v(t) = \begin{cases} 24(1 - e^{-t}) V & 0 < t < 1 \\ 30 - 14.83 e^{-(t-1)} V & t > 1 \end{cases}$$

⑥

⑥ $v(t) = 30(1 - e^{-t/2}) u(t) V$.

⑥
$$i(t) = \begin{cases} \frac{1}{6}(1 - e^{-t}) A & 0 < t < 1 \\ 0.5 - 0.39 e^{-(t-1)} A & t > 1 \end{cases}$$

⑥ $v(t) = 16 e^{-0.5t} V$.

⑥ $i_x(t) = 7.5(3 - e^{-4t}) mA$ $t > 0$

⑥ $i(t) = -9.6t e^{-20t} A$.

⑦ $C = 40mF$.

⑦ $i(t) = \left[e^{-4.43t} + e^{-0.903t} \right] A$.

⑦ $i(0) = 1$, $i(\infty) = \frac{6}{7}$
 $i(t) = \frac{1}{7}(6 - e^{-2t}) A$.

$$(73) \quad v_o(t) = 2(e^{-3t} - e^{-10t}) \text{ V.}$$

(2)

$$(74) \quad \text{overdamped.}$$

$$(75) \quad i(t) = (15 \cos 2t + 15 \sin 2t) e^{-2t} \text{ A.}$$

$$(76) \quad v(t) = \left[70 - (30 \cos 0.6t + 40 \sin 0.6t) e^{-0.8t} \right] \text{ V}$$

$$i(t) = 10 \sin 0.6t e^{-0.8t} \text{ A.}$$

(77)

$$(78) \quad i(t) = 2e^{-8t} u(t) \text{ A}$$

$$v(t) = -8e^{-8t} u(t) \text{ V.}$$

(82)

$$v_1(t) = 11.325 \cos(2t + 60.1^\circ) \text{ V}$$

$$v_2(t) = 33.02 \cos(2t + 57.12^\circ) \text{ V}$$

(88)

$$V_1 = 96.8 \angle 69.6^\circ \text{ V}$$

$$V_2 = 16.88 \angle 165.72^\circ \text{ V.}$$

(88)

$$I_o = 5.969 \angle 65.45^\circ \text{ A}$$

(92)

(94)

$$Z_{th} = 4.77 \angle 34.32^\circ \Omega$$

$$V_{th} = 9.61 \angle 64.55^\circ \text{ V}$$

$$Z_{total} = 5.97 \angle 6.626^\circ \Omega$$

$$I_1 = \frac{V_{th}}{Z_{total}} = 1.6 \angle 57.92^\circ \text{ A.}$$

(96) $v_o(t) = 3.835 \cos(4t - 35.02^\circ) \text{ V}$

(97) $Z_N = Z_{th} = 13.43 \angle 82.86^\circ \Omega$

$I_N = I_{sc} = 22 \angle 30^\circ \text{ A}$

ct flowing from a-b

$Z_{tot} = 14.26 \angle 52.57^\circ \Omega$, $V_{th} = 295.46 \angle 112.8^\circ \text{ V}$

$I = \frac{V_{th}}{Z_{tot}} = 20.72 \angle 60.3^\circ \text{ A}$

(100) $i = 4.47 \sin(200t + 56.56^\circ) \text{ A}$

(102) $I_o = 1.465 \angle 38.48^\circ \text{ A}$

(103) $Z_N = 3.176 + j 0.706 \Omega$

$I_N = 8.396 \angle -32.68^\circ \text{ A}$

$I_o = 1.9714 \angle -2.10^\circ \text{ A}$

(104) $v_o(t) = 10 + 21.45 \sin(2t + 26.56^\circ) + 10.73 \cos(3t - 26.56^\circ) \text{ V}$

(105) $I_1 = 381.4 \angle 109.6^\circ \text{ mA}$

$I_2 = 344.3 \angle 124.4^\circ \text{ mA}$

$I_3 = 145.5 \angle -60.42^\circ \text{ mA}$

$I_m = I_1 - I_2 = 100.5 \angle 48.5^\circ \text{ mA}$

(115) $y(t) = (2e^{-2t} - 2e^{-1.5t} \cos \sqrt{1.75}t + \frac{1}{\sqrt{1.75}} e^{-1.5t} \sin \sqrt{1.75}t) u(t)$

(120) $V_o = \frac{20}{s(s+3) + (s+5)}$

(121) a) $Z_{th} = 2s+3$ $V_o = \frac{5}{s+2} = \frac{125}{s(s+4)}$

b) $v_o(0) = 0$, $v_o(\infty) = 31.25 \text{ V}$ c) $v_o(t) = 31.25(1 - e^{-4t})u(t) \text{ V}$

$$(123) \quad V_0(s) = \frac{24(s+0.25)}{s(s+0.3)}$$

$$V_0(0) = 24 \text{ V.}$$

$$V_0(\infty) = 20 \text{ V.}$$

$$V_0(t) = (20 + 4e^{-0.3t})u(t) \text{ V}$$

$$(132) \quad L = \frac{R}{B} = 0.2387 \text{ H.}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$C = 2.653 \mu\text{F.}$$

$$Q = \frac{\omega_0}{B}$$

$$\omega_1 = \omega_0 - \frac{B}{2}$$

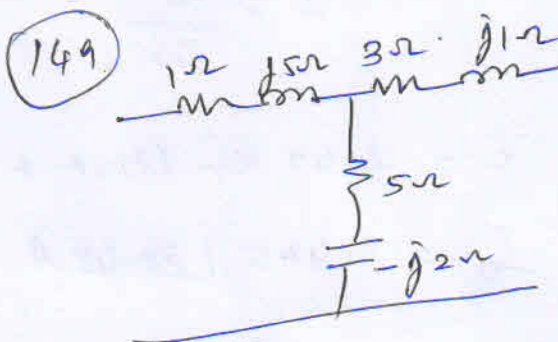
$$\omega_2 = \omega_0 + \frac{B}{2}$$

$$(139) \quad Z = \begin{bmatrix} 9.6 & -0.8 \\ -0.8 & 8.4 \end{bmatrix} \Omega.$$

$$(143) \quad Y = \begin{bmatrix} \frac{1}{8} & -\frac{1}{12} \\ -\frac{1}{12} & \frac{1}{2} \end{bmatrix} \text{ S.}$$

$$(146) \quad T = \begin{bmatrix} -3.5 & 0.833 \Omega \\ -2.5 \text{ S} & -0.5 \end{bmatrix}.$$

$$(148) \quad [Y] = [Y_a] + [Y_b] = \begin{bmatrix} 6+j2 & -4-j4 \\ -4-j4 & 7-j2 \end{bmatrix} \text{ S.}$$



$$(150) \quad I_1 = 5.5 I_2$$

$$\cancel{I_2 = 0.804}$$

$$I_1 = 5.5 I_2 = 0.8049 \text{ A}$$

$$V_2 = -10 I_2 = -1.46 \text{ V}$$

$$V_1 = 12 - 6 I_1 = 7.17 \text{ V}$$

$$I_2 = 0.1463 \text{ A.}$$